

Lab Session 1



Lab Session 1

1. Greeting Program

Write a program that asks the user for their name and age, then prints:

Hello [name], you are [age] years old!

2. Simple Calculator

Ask the user for two numbers. Print the sum of those numbers using **print()**.

3. Favorite Things

Take 3 inputs: favorite color, favorite food, and favorite hobby.

Print a sentence like: *My favorite color is blue, I love pizza, and I enjoy reading.*

4. Area of a Rectangle

Ask the user for the length and width of a rectangle.

Calculate and print the area with a message: *The area is [result].*

5. Name Length

Ask the user for their full name.

Print a message showing how many characters are in their name.

Hint: Use **len()** function.

6. Temperature Converter

Ask the user for temperature in Celsius.



Convert it to Fahrenheit using the formula $F = C \times \frac{9}{5} + 32$ and print the result.

7. Mad Libs Game

Ask for a noun, verb, and adjective.

Print a funny sentence like: *The [adjective] [noun] likes to [verb] all day.*

8. Age in Months

Ask the user for their age in years.

Print their approximate age in months.

Hint: **1 year = 12 months.**

9. Personalized Welcome

Ask for the user's first and last name separately.

Print a welcome message in 2 lines:

Welcome, [First Name]!

Your full name is [First Name] [Last Name].

10. Mini Bio Generator

Ask the user for: name, city, and favorite animal.

Print a short bio in one line:

Hi, I'm [name] from [city]. My favorite animal is [animal].

Lab Session 2



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Lab Session 2

Task 1: The Classic Welcome Banner

Objective: Use string multiplication to create a dynamic, scalable border around a welcome message.

```
=====
= WELCOME TO PYTHON PROGRAMMING LAB =
=====
```

Task 2: Digital Receipt Generator

Objective: Use f-string alignment specifiers (< for left, > for right) and dashes to format a clean retail receipt.

```
-----
|      RETAIL RECEIPT      |
-----
| Wireless Mouse           $ 25.50 |
| Mechanical Keyboard      $ 85.00 |
| USB-C Cable              $ 12.75 |
-----
| TOTAL:                   $ 123.25 |
-----
```

Task 3: Multi-Format Number Matrix

Objective: Display a single integer in decimal, binary, octal, and hexadecimal formats using f-string type codes inside a grid.

```
+-----+-----+-----+
| DECIMAL | BINARY | HEX |
+-----+-----+-----+
| 156     | 10011100 | 9C |
+-----+-----+-----+
```



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Task 4: Temperature Conversion Table

Objective: Calculate Celsius to Fahrenheit conversions and display them using a clean, tabular border structure.

Celsius (°C)	Fahrenheit (°F)
0.0	32.0
10.0	50.0
20.0	68.0
30.0	86.0
40.0	104.0

Task 5: Dynamic Multiplication Grid

Objective: Build a mini math matrix utilizing uniform widths to keep custom row/column dividers perfectly aligned.

MULTIPLICATION: 7	
7 x 1	7
7 x 2	14
7 x 3	21
7 x 4	28
7 x 5	35

Lab Session 3



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Lab Session 3

Task 1: The Contraction Crunch

- **Goal:** Print a string that contains a contraction.
- **Instruction:** Define a variable containing the sentence "I'm learning Python today!" without using any escape characters (like \).
- **Hint:** Wrap the entire string in the opposite quote type used by the contraction.

Task 2: Quote the Quote

- **Goal:** Print a direct quotation inside a sentence.
- **Instruction:** Store and print the phrase The programmer said, "Hello World", before coding. Keep the double quotes intact around Hello World.
- **Hint:** Use single quotes on the outside so the interior double quotes don't break the string.

Task 3: The Mixing Match

- **Goal:** Mix both quote types inside a single sentence.
- **Instruction:** Print the exact sentence: She said, "It's a beautiful day for coding."
- **Hint:** Because both and are inside the text, you must use an escape character (\'or \") for one of them, OR wrap the entire string in triple quotes (""" or...).

Task 4: Clean Up the Code Error

- **Goal:** Fix syntax errors caused by improper quoting.
- **Instruction:** The following code throws a SyntaxError. Rewrite it so it runs perfectly:
- `python`
- `# Fix this line:`
- `message = "My favorite book is "The Hobbit" because it's a classic."`
- **Use code with caution.**

Task 5: Multiline Matchup

- **Goal:** Create a multi-line block layout using quotes.



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- Instruction: Print the text below exactly as it looks, preserving the line breaks and the interior quotes:
` `` `
 - text
 - User Input: 'ASCII'
 - System Output: "01000001"
 - ` `` `
 - Use code with caution.
 - Hint: Use triple quotes ('''' or **') to span multiple lines cleanly without using \n.